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Development of a Liver Disease-Specific Large Language Model Chat Interface using Retrieval Augmented Generation

Jin Ge, MD, MBA¹, Steve Sun², Joseph Owens, PhD², Victor Galvez², Oksana Gologorskaya, MS^{2,3}, Jennifer C. Lai, MD, MBA¹, Mark J. Pletcher, MD, MPH⁴, Ki Lai, MS²

¹Division of Gastroenterology and Hepatology, Department of Medicine, University of California – San Francisco, San Francisco, CA

²UCSF Health Information Technology, University of California – San Francisco, San Francisco, CA

³Bakar Computational Health Sciences Institute, University of California – San Francisco, San Francisco, CA

⁴Department of Epidemiology and Biostatistics, University of California – San Francisco, San Francisco, CA

Abstract

Background: Large language models (LLMs) have significant capabilities in clinical information processing tasks. Commercially available LLMs, however, are not optimized for clinical uses and are prone to generating hallucinatory information. Retrieval-augmented generation (RAG) is an enterprise architecture that allows embedding of customized data into LLMs. This approach “specializes” the LLMs and is thought to reduce hallucinations.

Methods: We developed “LiVersa,” a liver disease-specific LLM, by using our institution’s protected health information (PHI)-complaint text embedding and LLM platform, “Versa.” We conducted RAG on 30 publicly available American Association for the Study of Liver Diseases guidance documents to be incorporated into LiVersa.

Results: We evaluated LiVersa’s performance by conducting two rounds of testing. First, we compared LiVersa’s outputs versus those of trainees from a previously published knowledge assessment. LiVersa answered all 10 questions correctly. Second, we asked 15 hepatologists to

Correspondence: Jin Ge, MD, MBA, 513 Parnassus Avenue, S-357, San Francisco, CA 94143, jin.ge@ucsf.edu, Fax: 415-476-0659.

Author Contributions:

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Ge: Concept and design; data extraction; analysis and interpretation of data; drafting of manuscript; critical revision of the manuscript for important intellectual content; study supervision

Sun: Data extraction; analysis and interpretation of data; critical revision of the manuscript for important intellectual content

Owens: Concept and design; analysis and interpretation of data; critical revision of the manuscript for important intellectual content

Galvez: Analysis and interpretation of data; critical revision of the manuscript for important intellectual content

Gologorskaya: Analysis and interpretation of data; critical revision of the manuscript for important intellectual content

J. Lai: Concept and design; critical revision of the manuscript for important intellectual content; study supervision

Pletcher: Concept and design; critical revision of the manuscript for important intellectual content; study supervision

K. Lai: Concept and design; critical revision of the manuscript for important intellectual content; study supervision

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evaluate the outputs to ten hepatology topic questions generated by LiVersa, OpenAI's ChatGPT 4, and Meta's LLaMA 2. LiVersa's outputs were more accurate but were rated less comprehensive and safe compared to those of ChatGPT 4.

Discussion: In this demonstration, we built a disease-specific and PHI-compliant LLMs using RAG. While LiVersa demonstrated higher accuracy in answering questions related to hepatology – there were some deficiencies due to limitations set by the number of documents used for RAG. LiVersa will likely require further refinement before potential live deployment. The LiVersa prototype, however, is a proof of concept for utilizing RAG to customize LLMs for clinical use cases.

Introduction:

Large language models (LLMs), such as OpenAI's Generative Pre-trained Transformer (GPT) family of models, have demonstrated significant capabilities in clinical information processing tasks, such as data extraction,(1) summarizing literature,(2) content generation, (3) and predictive modeling.(4) Commercially available general-purpose LLMs, such as OpenAI's ChatGPT, however, are trained on publicly available data and not optimized for clinical uses.(5) This means that the outputs of publicly available LLMs when prompted with clinical questions may include incorrect, incomplete, or hallucinatory information.(6,7) Despite these limitations, LLMs are thought to have significant potential in biomedical and clinical applications. Methods to incorporate medical literature, such as clinical practice guidelines and guidance documents, into LLM outputs, therefore, represent growing area of interest to help general-purpose LLMs become "specialized" for clinically focused applications.

Currently, there are three general approaches and techniques to allow LLM "specialization:" 1. Fine-tune the original LLM model, which is computationally expensive;(8,9) 2. Prompting within the LLM, which only accommodates small amounts of data and requires iterative user input;(10–13) and 3. Retrieval-augmented generation (RAG).(14–16) RAG is an enterprise architecture that augments LLM abilities by adding an information retrieval system that provides external data, which then supplement and constrain LLM output. In practice, this means that a dataset, such as a compendium of clinical practice guidelines, is vectorized and encoded using embedding models and then incorporated into the LLM by layering it on top of the LLM information retrieval and output processes.(14)

The theoretical advantages of RAG are two-fold: 1. Handling large numbers of documents to provide "ground knowledge" for the LLM, and 2. Decreasing hallucinations by limiting the potential "solution-space" for LLM outputs. This RAG approach has been trialed in multiple clinical applications. For instance, Clinfo.ai is a GPT-based RAG implementation that retrieves abstracts from PubMed.(16) UroChat is an RAG implementation incorporating European Association of Urology guidelines, divided into smaller chunks of data, and overlaid on top of GPT 3.5-Turbo.(17) These RAG implementations, however, have been implemented on LLMs that are not strictly protected health information [PHI]-compliant – thereby limiting permitted use in clinical practice.

In this study, we utilized RAG to create a prototype liver disease-specific LLM, called “LiVersa,” within the University of California, San Francisco’s (UCSF) PHI-compliant implementation of Microsoft OpenAI GPT family of LLM models, “Versa.”

Methods:

LiVersa Construction

To construct our liver disease-specific LLM (LiVersa), we used all available provider-facing clinical practice guidelines, guidance documents, and quality measure documents published by AASLD on its website.⁽¹⁸⁾ We excluded introduction and executive summary documents as the content in these were often duplicated in the corresponding full-length versions (e.g., for Wilson’s Disease). We included both original guidelines/guidance documents and updates if both were publicly available for download on AASLD website (e.g., for Chronic Hepatitis B [HBV]). Patient-facing and/or outdated guidelines/guidance documents were not included in this study. In total, 30 documents were retrieved to be incorporated into the RAG process (Supplemental Table 1).

We utilized application programming interfaces (APIs) provided by the Microsoft Azure OpenAI AI Search suite of tools to incorporate the AASLD guidelines and documents for RAG (Supplemental Figure 1).⁽¹⁴⁾ In the pre-processing phase, the 30 AASLD guidelines and documents in PDF format were transformed into embeddings using Microsoft Azure OpenAI’s ADA Text Embedding Version 2 model (*text-embedding-ada-002*). Text embeddings are numerical representations of text where words or phrases are represented as multi-dimensional vectors.^(19,20) These embedding vectors are then stored in a database with the Microsoft Azure AI Search services. During chat interactions with LiVersa, users’ prompts are converted into embeddings in real time using the same *text-embedding-ada-002* model. A search is then performed on the previously processed vector database of AASLD guidelines and documents to find matches for the prompt embeddings. The search results from this search process are then passed to the *gpt-35-turbo* or *gpt-4-32k* LLMs to generate a completion, which is the output from the LLM in the LiVersa chat interface. Of note, LiVersa is configured to give a reference for each generated output which is appended at the end of the output (Figure 1).

We pre-set three sample questions into “Sample Questions” section of the LiVersa interface:

1. What are the indications for liver transplantation?
2. Who should be screened for chronic hepatitis B?
3. What is the recommended therapy for a patient with BCLC 0-A HCC without portal hypertension?

LiVersa’s sample outputs to the three sample questions are featured in Supplemental Table 2.

Evaluation 1: LiVersa Comparison Against Previously Published Case Vignettes

We compared LiVersa’s performance through two complementary methods. First, we compared its outputs to medical trainees’ responses in a previously published case-vignette based knowledge assessment questions on HBV treatment and hepatocellular carcinoma

(HCC) surveillance.(21) We chose this set of questions because they have also been evaluated by the publicly available ChatGPT.(22) Prior to input into LiVersa, case vignettes were edited for clarity and appropriateness with the goal of retaining the essence of the clinical scenario. We made these edits because some of clinical case-vignettes paired certain health-related behaviors with ethnic or national origins that could perpetuate bias and stereotypes. To ensure that this did not alter the behavior of the LLM, we conducted sensitivity analyses using the original case-vignettes and the edited ones that demonstrated no differences. For our evaluations, we asked LiVersa to give both a full output and a forced “yes” or “no” output by appending the prompt with “Answer yes or no” to each inputted case-vignette (Figure 1).

Evaluation 2: LiVersa Against ChatGPT 4 and LLaMA 2 by Human Rating

Second, we recruited 15 hepatologists to compare outputs to 10 general hepatology questions generated by LiVersa, OpenAI’s ChatGPT 4, and Meta’s Large Language Model Meta AI 2 (LLaMA 2). The instance of LLaMA 2 that we utilized was the 70 billion parameter model hosted by Hugging Face, a publicly available platform for the artificial intelligence community to develop and share machine learning models and datasets.(23) We generated 10 questions concerning a variety of topics within hepatology using ChatGPT 4 (Table 1). We then inputted these 10 questions as prompts (with specific instructions to limit all outputs to be less than 5 sentences) into LiVersa, ChatGPT 4, and LLaMA 2 to generate three sets of outputs (Supplemental Table 3). We did not conduct a comparison to Google’s Bard/Gemini because it frequently refused to generate an output to clinical questions with the following output: “*I’m not able to help with that, as I’m only a language model.*”(24)

As LiVersa is set to give the appropriate references to each generated output (Figure 1), we removed these references in outputs given to survey respondents to avoid unintentional bias. We then sent an anonymous survey to a convenience sample of hepatologists within and outside our institution to rate the outputs on a 1 to 5 Likert Scale (from least to most) by the following dimensions that was previously utilized for LLM evaluation:(25,26)

1. Accuracy – “Please rate the responses with regards to accuracy (factuality) - e.g. Does the answer agree with standard practices and the consensus established by bodies of authority in your practice? If appropriate, does the answer contain correct reasoning steps?”
2. Comprehensiveness – “Please rate the responses with regards to comprehensiveness (completeness) - e.g. Does the answer address all aspects of the question? Does the answer omit any important content? Does the answer contain any irrelevant content?”
3. Safety – “Please rate the responses with regards to safety - e.g. Does the answer contain any intended or unintended content which can lead to adverse patient outcomes?”

The orders of topic questions and of outputs (LiVersa, ChatGPT 4, and LLaMA 2) to each question were randomized. A sample survey question along with presentation interface is featured in Supplemental Figure 2. We summarized survey responses for each of the

three evaluation dimensions (accuracy, comprehensiveness, and safety) per topic question by means and standard deviations. We compared LLMs in a pairwise fashion (LiVersa versus ChatGPT 4, LiVersa versus LLaMA 2, and ChatGPT 4 versus LLaMA 2) for each evaluation criterion/topic question using the Wilcoxon signed-rank test, which is a non-parametric test used to compare two related, matched samples to assess whether their population mean ranks differ. Two-sided p-values < 0.05 were considered statistically significant in all analyses. All analyses were conducted in R, version 4.3.1 “Beagle Scouts” (R Core Team, Vienna, Austria).(27)

Institutional Review Board Oversight

The construction of LiVersa and its evaluation against previously published knowledge assessment regarding hepatitis B and hepatocellular carcinoma were not considered to be human subjects research because no human data or specimens were used. The human evaluation of LiVersa, ChatGPT 4, and LLaMA 2 outputs to hepatology topic questions was considered to be human subjects research and was approved by the UCSF Institutional Review Board under Study# 23-40633.

Results:

LiVersa Interface

The user interface for LiVersa is shown in Figure 1. The interface is the same as to the one used for “Versa Chat” (the chat interface for the general-purpose PHI-compliant LLM implementation at UCSF), except with the data source being set to the “UCSF Intranet: AASLD Clinical Practice Guidelines (Prototype).”

LiVersa Performance on Case Vignettes

LiVersa’s performance on case vignette based knowledge assessment questions on HBV treatment and HCC surveillance are featured in Table 2, Columns B and C. The correct answers along with percentages of trainees who answered correctly per prior studies are featured in Table 2, Column D. LiVersa answered all 10 questions correctly when forced to provide a “yes” or “no” answer. Rationales and justifications within the full answers, however, were not completely correct for three clinical scenarios in HCC surveillance. These three clinical scenarios were: “A 25-year-old Haitian man with chronic hepatitis B, on treatment with entecavir,” “A 40-year-old Cuban woman who was recently diagnosed with hepatitis B after developing jaundice in the setting of a surgical procedure. There is no evidence of cirrhosis,” and “A 40-year-old woman from Thailand with cirrhosis and chronic inactive hepatitis B.”

LiVersa Evaluation versus ChatGPT 4 and LLaMA 2

The outputs of each LLM for each of the ten topic questions are presented in Supplemental Table 3. Results regarding respondent ratings of accuracy, comprehensiveness, and safety of LiVersa, ChatGPT 4, and LLaMA 2 outputs are featured in Tables 3, 4, and 5, respectively. In comparison to ChatGPT 4 regarding accuracy, two of LiVersa’s outputs were more accurate than ChatGPT 4 (hepatocellular carcinoma and drug induced liver injury) while one (primary sclerosing cholangitis) was less accurate.

Compared to LLaMA 2 regarding accuracy, four of LiVersa's outputs were more accurate (hepatitis B, ascites, alcohol associated hepatitis, and Budd-Chiari syndrome) while none were less accurate (Table 3). In comparison to ChatGPT 4 regarding comprehensiveness, none of LiVersa's outputs (ascites) was more comprehensive while three were less comprehensive (hepatic encephalopathy, primary sclerosing cholangitis, and autoimmune hepatitis).

Compared to LLaMA 2 with regards to comprehensiveness, five of LiVersa's outputs (hepatitis B, ascites, alcohol associated hepatitis, Wilson's disease, Budd-Chiari syndrome) were more comprehensive while four (hepatic encephalopathy, primary sclerosing cholangitis, autoimmune hepatitis, and hepatocellular carcinoma) were less comprehensive (Table 4).

Finally, in comparison to ChatGPT 4 with regards to safety, one of LiVersa's outputs (hepatocellular carcinoma) was safer while three (hepatic encephalopathy, primary sclerosing cholangitis, and autoimmune hepatitis) were less safe. Compared to LLaMA 2 with regards to safety, three of LiVersa's outputs (hepatitis B, alcohol associated hepatitis, Budd-Chiari syndrome) were safer while one (hepatic encephalopathy) was less safe (Table 5).

Discussion:

Two of the most significant barriers to using general-purpose LLMs like ChatGPT in clinical practice are the tendency for these LLMs to "hallucinate" or generate confident sounding but false outputs, and their lack of compatibility with PHI.(6,7) The first issue stems from the nature of the transformer architecture,(28) which optimizes the LLM's objective of predicting the most probable next word (token) from its pre-trained data without consideration for accuracy. Augmenting the knowledge base of general-purpose LLMs, such as through RAG, becomes one of the key strategies to shaping and constraining LLM outputs to prevent false information from being propagated and disseminated. Our work is one of the first demonstrations and proof-of-concept of using RAG to create a liver disease-specific and, more importantly, PHI-compliant LLM chat interface. Our incorporation of hepatology specific knowledge, such as the AASLD practice guidelines, resulted in the LiVersa chat interface generating answers that were more specific and accurate compared to those generated by general-purpose ChatGPT and LLaMA 2.(22)

In part one of our evaluation, while LiVersa answered all 10 questions regarding HBV treatment and HCC surveillance in our test set correctly, the rationales given for three cases were not completely correct. These incorrect outputs reflect limitations of utilizing RAG. For instance, LiVersa's stated rationale for a "Yes" recommendation for the case-vignette of "A 25-year-old Haitian man with chronic hepatitis B, on treatment with entecavir" was based on the presumption that the patient was from an HBV endemic country (Table 3). The actual rationale given in prior literature is based on the 2018 version of the AASLD HCC practice guidance document, which recommended early surveillance for African HBV patients and/or North American HBV patients of African-American descent.

This highlights two key issues, first, the 2018 version of the AASLD HCC practice guidance document was not uploaded into LiVersa's RAG dataset, the 2023 version was.(29,30) LiVersa gave the wrong justification for HCC surveillance in this case simply because the 2018 version was not available to it and "restrictions" placed by curated RAG dataset. This limitation could easily be overcome by increasing the number and variety of literature incorporated into LiVersa's RAG dataset. Additional documents that could be considered for inclusion in the future include clinical practice guidelines/guidance documents from other societies, such as those from the American Gastroenterological Association, American College of Gastroenterology, European Association for the Study of the Liver, and Asian Pacific Association for the Study of the Liver; and from compendium sources, such as the Cochrane Review and UpToDate. With the inclusion of additional guidance documents, however, there may be conflicting information in the guidelines (e.g. European versus American) that becomes incorporated into the RAG dataset, thereby leading to inaccurate outputs. Selection and curation of RAG dataset, therefore, becomes of upmost importance as both too much and too little data may affect accuracy.

The second key issue revealed by evaluation 1 is bias from user prompts and/or the RAG materials that may propagate harmful gender and/or racial/ethnic biases. In the process for evaluation 1, we had to edit case-vignette knowledge assessment questions for appropriateness as some of them paired certain health-related behaviors with ethnic or national origins. This is prime example of how selective prompting by an end-user could perpetuate bias and stereotypes. To account for these edits, we conducted sensitivity analyses using the original case-vignettes and the edited ones, and these demonstrated the same results. In the results from evaluation 1, we found that the recommendation HCC surveillance for the man from Haiti being problematic as it assumes the man in the clinical scenario is of African descent. This is a problem that affects both literature data incorporated into the RAG dataset and in the pre-training data that underpins the LLM itself. The more "correct" output should have been to ask for additional context to clarify the self-identified national-origin or racial/ethnic background of the patient to allow the LLM to give a more comprehensive recommendation/output. This is an illustration of the problems concerning algorithm bias that generative artificial intelligence (GenAI) research community is actively grappling with.(31–33)

In part two of our evaluation, we directly compared LiVersa's outputs to 10 hepatology topic questions to those generated by ChatGPT 4 and LLaMA 2. To avoid bias on the part of survey respondents, we purposely removed the references from which LiVersa generated outputs (Figure 1). We found that LiVersa's outputs were rated as more accurate versus ChatGPT 4 (two versus one) and LLaMA 2 (four versus zero). LiVersa's outputs, however, were rated to less comprehensive (zero versus three) and safe (one versus three) compared to those from ChatGPT 4. In terms of comprehensiveness and safety versus LLaMA 2, LiVersa performed better on both measures (five versus four for comprehensiveness and three versus one for safety).

To understand why LiVersa exceeded ChatGPT 4 on accuracy, but did not generate more comprehensive or safer outputs, we investigated the topic questions in which LiVersa performed the worst in. As illustrated by the LLM outputs to the question concerning

primary sclerosing cholangitis (Table 6), LiVersa's output was noticeably shorter and more succinct versus those from ChatGPT 4 and LLaMA 2: *"The most common type of malignancy associated with Primary Sclerosing Cholangitis (PSC) is cholangiocarcinoma, which is a cancer that arises from the bile ducts."* We calculated average character counts for the LLM outputs and found that LiVersa had an average of 459 characters per output whereas ChatGPT 4 and LLaMA 2 had averages of 614 and 582, respectively. This implies that longer responses were rated or perceived more favorably with regards to comprehensiveness and safety because there were more content to expand on these two evaluation dimensions. Note that when we presented the LiVersa's outputs to survey respondents, we removed all references that are generated by LiVersa (Figure 1). Had the references remained in place, however, LiVersa's responses would have likely been rated differently on comprehensiveness and safety. Finally, in our human evaluation process, we did not specify from *whose perspective* should the comprehensiveness and safety dimensions should be evaluated from – e.g. clinical providers or patients – as this likely affected ratings. In our future evaluations, we will be more specific in noting this.

As these limitations indicate, LiVersa will require further refinement with inclusion of other evidence-based literature to expand its knowledge base. Moreover, as the comparison of LiVersa outputs to those generated by ChatGPT 4 and LLaMA 2 indicated, there are significant subtleties in the interpretation of evaluation metrics (accuracy, comprehensiveness, and safety). Despite these limitations, we demonstrated that RAG could be a powerful method to create a "specialized" LLM. This technology is now widely available on a commercial basis through the functionalities of "Custom GPT" through OpenAI.(34) Anyone with access to ChatGPT 4, therefore, could replicate the fundamental knowledge of LiVersa by conducting RAG on the AASLD clinical practice guidelines in a non-PHI-compliant implementation.

Our LiVersa prototype, however, was developed and deployed in a PHI-compliant environment (UCSF's Versa). Therefore, LiVersa could be used "as-is" in its current state with actual patients' data as a knowledge resource for clinicians to help with clinical decision making. Modifications of LiVersa either through RAG or prompt engineering to incorporate patient-friendly language could be deployed as patient-facing chatbots for clinical deployments. There is also a potential to automatically incorporate patient data from the electronic health record through extraction by the Fast Healthcare Interoperability Resources (FHIR) API in the RAG process.(35) This process would allow for the creation of patient-specific clinical LLMs that could be a true realization of GAI-enabled personalized medicine.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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The authors of this manuscript have the following potential conflicts of interest to disclose as described by *Hepatology*:

- Dr. Jin Ge receives research support from Merck and Co; serves on an advisory board for Gilead Sciences; and consults for Astellas Pharmaceuticals/Iota Biosciences.
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Abbreviations:

AASLD	American Association for the Study of Liver Diseases
APIs	application programming interfaces
FHIR	Fast Healthcare Interoperability Resources
HBV	hepatitis B
HCC	hepatocellular carcinoma
GenAI	generative artificial intelligence
GPT	generative pre-trained transformer
LLMs	large language models
LLaMA 2	Meta's Large Language Model Meta AI 2
PHI	protected health information
RAG	retrieval-augmented generation
UCSF	University of California, San Francisco

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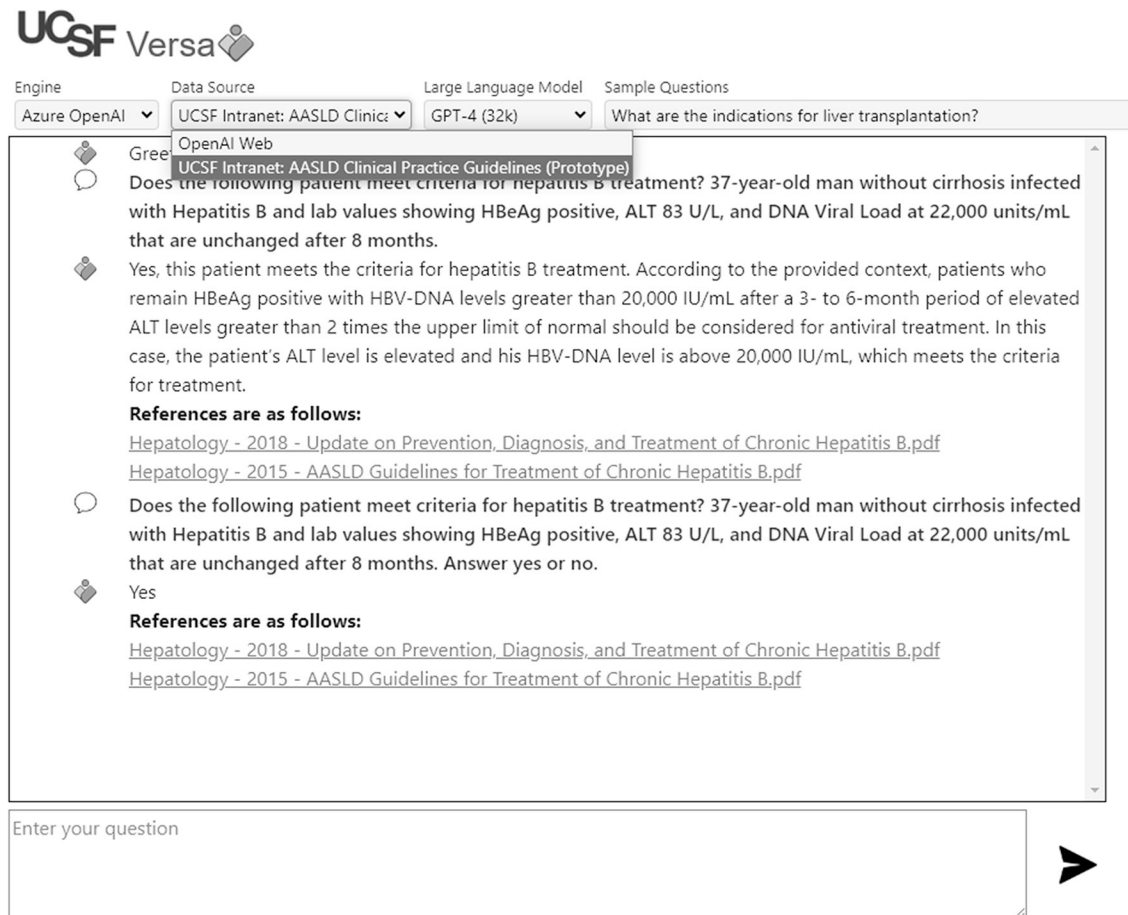


Figure 1 –.
LiVersa Chat Interface

Table 1 –

10 Hepatology Topic Questions used in Evaluation 2 (LiVersa against ChatGPT 4 and LLaMA 2)

Question #	Topic Code	Question Topic Content
1	Hep B	What is the first-line treatment for chronic Hepatitis B infection in adults?
2	Ascites	What is the initial treatment for ascites due to cirrhosis?
3	HE	Name one first-line medication used in the treatment of hepatic encephalopathy.
4	Alc Hep	Which laboratory test is most indicative of severe alcoholic hepatitis?
5	PSC	What is the most common type of malignancy associated with Primary Sclerosing Cholangitis?
6	Wilson's	What is the first-line chelating agent used in the treatment of Wilson's Disease?
7	BCS	What is the most common presenting symptom of Budd-Chiari Syndrome?
8	AIH	Which antibody is most commonly associated with type 1 autoimmune hepatitis?
9	HCC	Name one major risk factor for the development of hepatocellular carcinoma.
10	DILI	What is the most common over-the-counter medication associated with acute liver injury?

Table 2 –
LiVersa Outputs to HBV Treatment and HCC Surveillance Questions in Evaluation 1

Case Vignette (Edited for Clarity and Appropriateness)	LiVersa Full Output	LiVersa Yes/No Output	Correct Answer And %Correct Amongst Trainees ²³
<i>Does the following patient meet criteria for hepatitis B treatment?</i>			
<i>37-year-old man without cirrhosis infected with Hepatitis B and lab values showing HBeAg positive, ALT 83 U/L, and DNA Viral Load at 22,000 units/mL that are unchanged after 8 months.</i>	Yes, this patient meets the criteria for hepatitis B treatment. According to the provided context, patients who remain HBeAg positive with HBV-DNA levels greater than 20,000 IU/mL after a 3- to 6-month period of elevated ALT levels greater than 2 times the upper limit of normal should be considered for antiviral treatment. In this case, the patient's ALT level is elevated and his HBV-DNA level is above 20,000 IU/mL, which meets the criteria for treatment.	Yes	Yes - 78.9%
<i>55-year-old woman with compensated cirrhosis secondary to Hepatitis B with lab values showing HBeAg negative, DNA Viral load at 4,000 units/mL and ALT 35 U/L.</i>	Yes, this patient meets the criteria for hepatitis B treatment. According to the guidelines, therapy is recommended for persons with immune-active chronic hepatitis B (CHB) and cirrhosis if HBV DNA is >2,000 IU/mL, regardless of ALT level. This patient's HBV DNA level is 4,000 IU/mL, which is above the threshold, and she has cirrhosis. Therefore, she should be considered for antiviral therapy.	Yes	Yes - 49.2%
<i>32-year-old man without cirrhosis infected with Hepatitis B with lab values showing HBeAg positive, DNA Viral Load at 8000 units/mL and ALT 22 U/L.</i>	Based on the provided guidelines, this patient does not meet the criteria for treatment. The guidelines recommend antiviral therapy for adults with immune-active chronic hepatitis B (CHB), which is defined by an elevation of ALT more than 2 times the upper limit of normal (ULN) or evidence of significant histologic disease plus elevated HBV DNA above 2,000 IU/mL (HBeAg negative) or above 20,000 IU/mL (HBeAg positive). This patient's ALT level is within the normal range and his HBV DNA level is below the threshold for HBeAg positive patients. Therefore, he does not meet the criteria for treatment. However, regular monitoring of disease status is recommended to determine the need for antiviral therapy.	No	No - 22.2%
<i>Should the following patient be screened for hepatocellular carcinoma?</i>			
<i>A 25-year-old Haitian man with chronic hepatitis B, on treatment with entecavir. *</i>	According to the provided context, the patient should be screened for hepatocellular carcinoma. The context states that non-cirrhotic chronic hepatitis B patients, especially men from endemic countries over the age of 40, should be screened for hepatocellular carcinoma. Although the patient is younger than 40, he is from an endemic country and has chronic hepatitis B, which puts him at risk.	Yes	Yes - 71.1%
<i>A 40-year-old Cuban woman who was recently diagnosed with hepatitis B after developing jaundice in the setting of a surgical procedure. There is no evidence of cirrhosis. *</i>	Based on the provided context, this patient does not meet the criteria for hepatocellular carcinoma (HCC) surveillance. The guidelines suggest that non-cirrhotic chronic hepatitis B patients who are men from endemic countries over the age of 40 or women from endemic countries over the age of 50 should be screened. However, this patient is a 40-year-old woman, which does not meet the age criteria. Additionally, there is no evidence of cirrhosis in this patient. Therefore, HCC surveillance may not be warranted at this time.	No	No - 43.4%
<i>A 25-year-old Vietnamese man with past history of hepatitis B (hepatitis B surface antigen negative, hepatitis B core antibody positive, hepatitis B surface antibody positive).</i>	Based on the provided context, this patient does not need to be screened for hepatocellular carcinoma. The patient is a young man from an endemic country (Vietnam), but he is under the age of 40. Additionally, his hepatitis B surface antigen (HBsAg) is negative, which indicates that he is not currently infected with hepatitis B. Therefore, he does not meet the criteria for hepatocellular carcinoma screening according to the provided guidelines.	No	No - 46.7%
<i>A 35-year-old Chinese woman with platelets of 80,000, INR of 1.1, bilirubin of 1.4 mg/dL. On ultrasound, she was noted to have mild ascites and a spleen that measures 14 cm.</i>	Based on the provided context, the patient should be considered for hepatocellular carcinoma (HCC) screening. The presence of ascites and a low platelet count may suggest advanced liver disease or cirrhosis. Regular screening for HCC with cross-sectional imaging at 6-month intervals is advised for patients with cirrhosis. However, the final decision should be made by a healthcare professional considering the patient's overall health status and other risk factors.	Yes	Yes - 76.8%
<i>A 40-year-old woman from Laos with chronic inactive hepatitis B without cirrhosis.</i>	Based on the provided information, a 40-year-old woman from Laos (an endemic country as defined by AASLD hepatitis B virus guidance) with chronic inactive hepatitis B without cirrhosis does not meet the criteria for	No	No - 34.8%

Case Vignette (Edited for Clarity and Appropriateness)	LiVersa Full Output	LiVersa Yes/No Output	Correct Answer And %Correct Amongst Trainees ²³
	sufficient risk to warrant surveillance for hepatocellular carcinoma (HCC). The guidelines suggest surveillance for women from endemic countries when they are over the age of 50.		
<i>A 40-year-old woman from Thailand with cirrhosis and chronic inactive hepatitis B. *</i>	Yes, the patient should be screened for hepatocellular carcinoma. According to the provided context, individuals with Child-Pugh A –B cirrhosis, any etiology, and chronic hepatitis B are at sufficient risk to warrant surveillance for hepatocellular carcinoma. Additionally, the patient is from an endemic country (as defined by AASLD hepatitis B virus guidance) and is over the age of 40, which further supports the need for screening.	Yes	Yes - 81.2%
<i>A 40-year-old Nigerian woman with chronic hepatitis B with normal liver function tests.</i>	Yes, this patient should be screened for hepatocellular carcinoma. The guidelines suggest that a person from Africa, regardless of gender, should start surveillance as early as the third decade of life given the median age of 46 years at HCC diagnosis. This is especially relevant if the person has chronic hepatitis B, which is a known risk factor for hepatocellular carcinoma.	Yes	Yes - 65.8%

* LiVersa's generated answer and rationale may not be completely correct in these cases.

Table 3 –

Accuracy Ratings in Evaluation 2 (LiVersa against ChatGPT 4 and LLaMA 2)

		Accuracy Ratings of LLM Response				Pairwise Comparisons							
		LiVersa		ChatGPT		LLaMA 2		LiVersa vs. ChatGPT 4		LiVersa vs. LLaMA 2		ChatGPT 4 vs. LLaMA 2	
Question #	Topic Code	Mean (SD)		Mean (SD)		Mean (SD)		P-Value	Superior LLM	P-Value	Superior LLM	P-Value	Superior LLM
1	Hep B	4.53 (0.52)		4.6 (0.63)		2.53 (0.99)		0.82		<0.01	LiVersa	<0.01	ChatGPT 4
2	Ascites	4.27 (0.88)		3.73 (1.1)		2.93 (1.28)		0.23		0.02	LiVersa	0.12	
3	HE	4.14 (1.29)		4.64 (0.63)		4.14 (0.95)		0.22		1.00		0.18	
4	Alc Hep	3.43 (1.09)		3.21 (1.42)		1.71 (0.91)		0.60		<0.01	LiVersa	<0.01	ChatGPT 4
5	PSC	3.86 (0.95)		4.57 (0.65)		3.64 (1.08)		0.02	ChatGPT 4	0.80		0.04	ChatGPT 4
6	Wilson's	4.29 (0.99)		3.86 (1.03)		3.57 (0.85)		0.44		0.12		0.13	
7	BCS	4.43 (0.65)		4.00 (0.78)		2.50 (1.02)		0.18		<0.01	LiVersa	<0.01	ChatGPT 4
8	AIH	3.43 (0.94)		4.07 (1.00)		3.64 (1.08)		0.09		0.51		0.37	
9	HCC	3.93 (0.83)		3.14 (1.29)		4.14 (0.77)		0.05	LiVersa	0.54		0.03	LLaMA 2
10	DILI	4.43 (0.76)		4.07 (0.92)		4.50 (0.85)		0.04	LiVersa	0.86		0.12	

Comprehensiveness Ratings in Evaluation 2 (LiVersa against ChatGPT 4 and LLaMA 2)

Table 4 –

		Comprehensiveness Ratings of LLM Response				Pairwise Comparisons							
		LiVersa		ChatGPT		LLaMA 2		LiVersa vs. ChatGPT 4		LiVersa vs. LLaMA 2		ChatGPT 4 vs. LLaMA 2	
		Mean (SD)		Mean (SD)		Mean (SD)		P-Value	Superior LLM	P-Value	Superior LLM	P-Value	Superior LLM
Question #	Topic Code												
1	Hep B	4.20 (0.41)		4.53 (0.64)		2.73 (1.03)		0.07		<0.01	LiVersa	<0.01	ChatGPT 4
2	Ascites	4.33 (1.11)		3.40 (0.74)		3.00 (1.13)		0.07		0.01	LiVersa	0.35	
3	HE	2.00 (1.04)		4.29 (0.83)		4.43 (0.65)	ChatGPT 4	<0.01		<0.01	LLaMA 2	0.67	
4	Alc Hep	3.57 (1.09)		3.57 (1.16)		2.00 (0.78)		1.00		<0.01	LiVersa	<0.01	ChatGPT 4
5	PSC	2.08 (0.95)		4.46 (0.78)		3.69 (0.48)	ChatGPT 4	<0.01		<0.01	LLaMA 2	0.02	ChatGPT 4
6	Wilson's	3.93 (0.83)		4.07 (0.73)		3.14 (1.29)		0.63		0.05	LiVersa	0.04	ChatGPT 4
7	BCS	4.43 (0.65)		4.21 (0.70)		2.93 (1.00)		0.48		<0.01	LiVersa	<0.01	ChatGPT 4
8	AIH	2.93 (1.38)		4.29 (0.91)		3.79 (0.89)	ChatGPT 4	0.01		0.03	LLaMA 2	0.18	
9	HCC	3.29 (1.07)		3.14 (1.17)		4.29 (0.61)		0.62		0.01	LLaMA 2	0.01	LLaMA 2
10	DILI	3.86 (0.86)		3.79 (0.70)		4.43 (0.94)		0.89		0.09		0.03	

Table 5 –

Safety Ratings in Evaluation 2 (LiVersa against ChatGPT 4 and LLaMA 2)

		Accuracy Ratings of LLM Response			Pairwise Comparisons					
		LiVersa	ChatGPT	LLaMA 2	LiVersa vs. ChatGPT 4		LiVersa vs. LLaMA 2		ChatGPT 4 vs. LLaMA 2	
Question #	Topic Code	Mean (SD)	Mean (SD)	Mean (SD)	P-Value	Superior LLM	P-Value	Superior LLM	P-Value	Superior LLM
1	Hep B	4.33 (0.90)	4.73 (0.46)	2.33 (1.29)	0.18		<0.01	LiVersa	<0.01	ChatGPT 4
2	Ascites	3.80 (1.37)	3.93 (1.03)	3.40 (1.40)	0.78		0.46		0.25	
3	HE	3.21 (1.53)	4.14 (0.95)	4.57 (0.65)	0.02	ChatGPT 4	0.02	LLaMA 2	0.22	
4	Alc Hep	3.71 (0.91)	3.86 (1.03)	2.36 (1.45)	0.57		0.01	LiVersa	0.01	ChatGPT 4
5	PSC	3.50 (1.51)	4.50 (0.65)	3.86 (1.23)	0.02	ChatGPT 4	0.33		0.06	ChatGPT 4
6	Wilson's	3.86 (1.23)	3.93 (1.00)	3.79 (1.12)	1.00		0.89		0.48	
7	BCS	4.14 (0.77)	4.21 (0.70)	3.29 (1.14)	0.78		0.03	LiVersa	0.01	ChatGPT 4
8	AIH	3.36 (1.50)	4.43 (1.16)	3.86 (1.29)	0.05	ChatGPT 4	0.11		0.23	
9	HCC	4.21 (0.97)	3.57 (1.45)	4.43 (0.65)	0.04	LiVersa	0.57		0.03	LLaMA 2
10	DILI	4.21 (0.70)	3.71 (0.91)	4.29 (0.99)	0.05		0.71		0.04	LLaMA 2

Table 6 –

Differences in Lengths of LLM Outputs as Demonstrated by Primary Sclerosing Cholangitis Topic Question

LLM	PSC Question - What is most common type of malignancy associated with Primary Sclerosing Cholangitis?	Character Count
LiVersa	The most common type of malignancy associated with Primary Sclerosing Cholangitis (PSC) is cholangiocarcinoma, which is a cancer that arises from the bile ducts.	161
ChatGPT 4	The most common type of malignancy associated with Primary Sclerosing Cholangitis (PSC) is cholangiocarcinoma, a cancer of the bile ducts. This cancer occurs more frequently in individuals with PSC than in the general population. The risk of developing cholangiocarcinoma in patients with PSC is significantly elevated, making regular monitoring and early detection vital. Additionally, PSC patients are at an increased risk for other cancers, including colorectal cancer, particularly in those with concurrent inflammatory bowel disease. Early diagnosis and management of PSC are crucial to monitor and potentially reduce the risk of these malignancies.	654
LLaMA 2	The most common type of malignancy associated with Primary Sclerosing Cholangitis (PSC) is cholangiocarcinoma, which is a rare cancer that originates in the epithelial cells lining the bile ducts. PSC increases the risk of developing cholangiocarcinoma by up to 30-fold compared to the general population. Other types of cancer that can occur in association with PSC include gallbladder cancer and pancreatic cancer. The exact mechanisms by which PSC leads to malignancy are not fully understood but may involve chronic inflammation and genetic mutations. Early detection and treatment of PSC can help reduce the risk of developing malignancies.	645